# AI Driven Automotive Predictive Maintenance System

**Group Number: 31**





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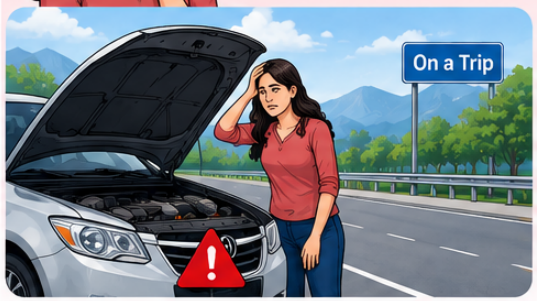

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# Introduction

## Reactive Maintenance
(Fix after failure)



"I don't check my car regularly. I only take it to the mechanic when something goes wrong."




- Hidden issues may go undetected even after a quick check.
- Car may break down unexpectedly (e.g., during a trip).
- High and unplanned repair cost.
- Inconvenience and time loss.

## Preventive Maintenance
(Service at fixed intervals)



"I take my car to the showroom for regular servicing. I get every part checked or replaced, even if it may not be necessary."




- Higher cost due to unnecessary servicing and premature replacement of healthy components.
- Time-consuming (scheduling, waiting).
- Resource and part wastage.
- Lower chance of breakdown, but not cost-efficient.

## Predictive Maintenance
(Maintain based on actual condition)



"I have a mobile app that shows the real-time condition and trends of my car's sensors, my driving behaviour, and alerts me about potential issues."




- Early detection of abnormal behaviour and probable faults.
- Maintenance only when needed (targeted and cost-efficient).
- Saves time, money and resources.
- More reliable, safer and smoother driving experience.

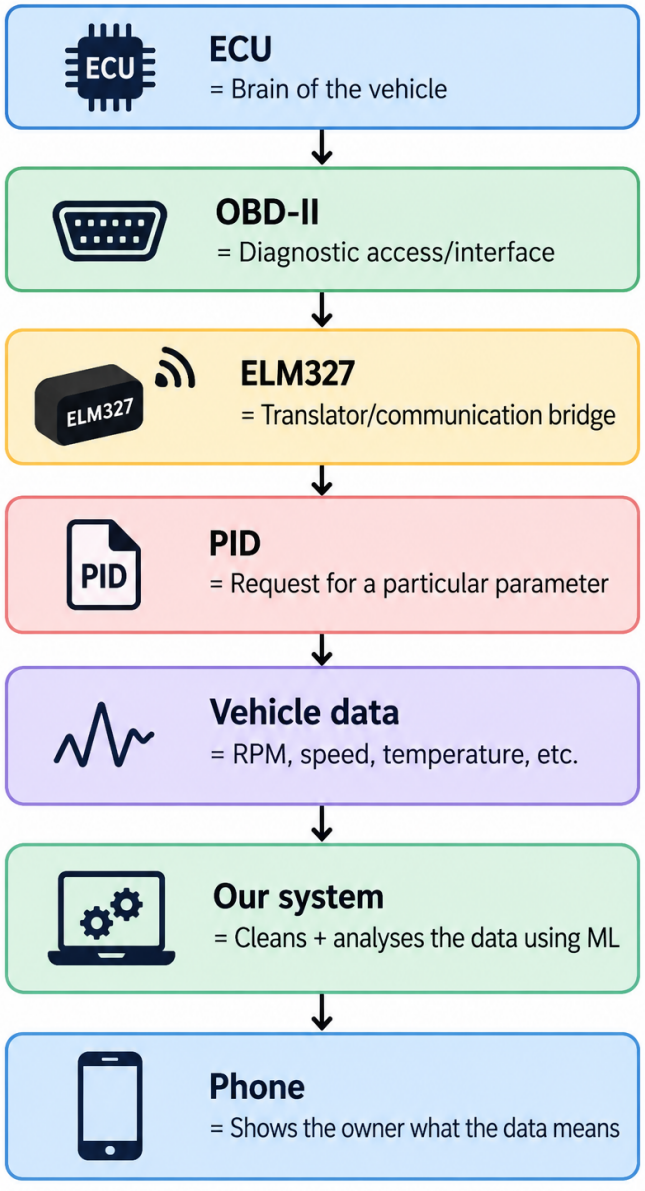
**Reactive**
Breakdown risk + unexpected cost

**Preventive**
Unnecessary maintenance + time and resource wastage

**Predictive**
Early detection + targeted maintenance + reduced cost and downtime

**From reacting to failures → to data-driven, smarter maintenance.**

# Introduction



# Introduction

- Vehicles generate rich sensor data, but it remains too technical for drivers to interpret.
- Issues are usually detected only **after** a breakdown occurs.
- Driving behavior also affects vehicle health, but this link is rarely communicated.
- **Goal:** Enable proactive, easy-to-understand vehicle health monitoring.

# Objectives

- Collect and study relevant vehicle sensor datasets for analysis. Preprocess and structure data for model training .Chose the best model to detect abnormal patterns in the data. Classify probable fault types from detected anomalies. Study the link between driving behavior and vehicle wear. Design a system to present findings in a clear, understandable format.

*Review 1: Dataset identification, objective finalization, and literature survey, preprocessing and chose the best Model for each application.*

# Block Diagram

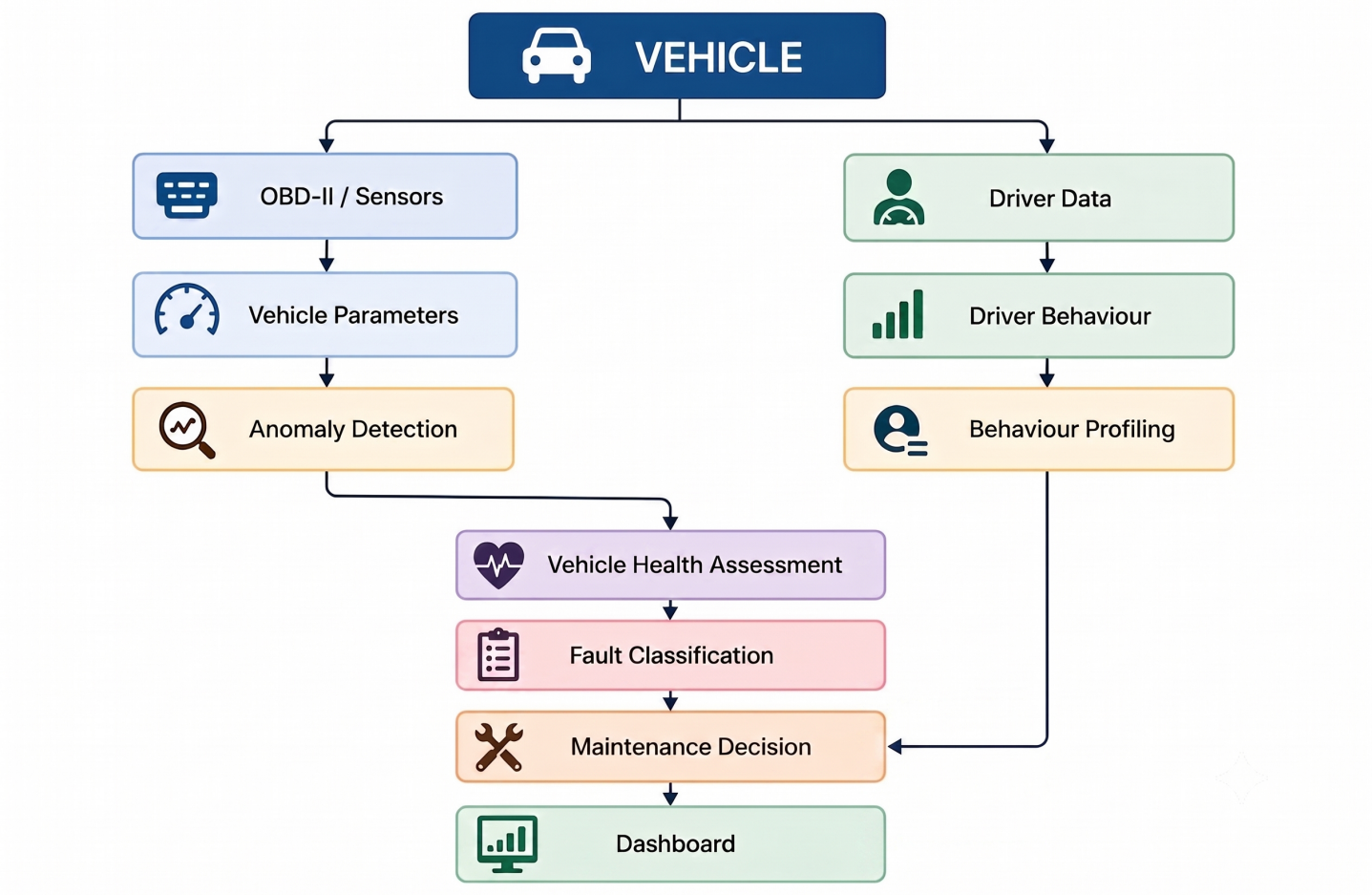



Fig 1. Overall block diagram of the proposed predictive maintenance system

# Sustainable Development Goal's Map

| SDG | Relevance | Contribution |
|---|---|---|
| **SDG 9 – Industry, Innovation and Infrastructure** | High | Applies AI/ML, OBD-II/sensor data and cloud technologies for intelligent vehicle maintenance. |
| **SDG 12 – Responsible Consumption and Production** | High | Supports condition-based maintenance and can reduce unnecessary component replacement and resource use. |
| **SDG 11 – Sustainable Cities and Communities** | Moderate | Supports safer and more reliable transportation through early detection of vehicle abnormalities and faults. |

# Literature Survey

| S.No. | Title | Authors | Conference / Journal | Year |
|---|---|---|---|---|
| 1 | A Review of OBD-II-Based Machine Learning Applications for Sustainable, Efficient, Secure, and Safe Vehicle Driving | E. T. Michailidis, A. Panagiotopoulou, A. Papadakis | *Sensors* (MDPI) | 2025 |
| 2 | Driving behavior analysis and classification by vehicle OBD data using machine learning | R. Kumar, A. Jain | *The Journal of Supercomputing* (Springer) | 2023 |
| 3 | Time-Series Anomaly Detection in Automated Vehicles Using D-CNN-LSTM Autoencoder | F. Khanmohammadi, R. Azmi | *IEEE Transactions on Intelligent Transportation Systems* | 2024 |
| 4 | XGBoost Method-Based Gearbox Fault Diagnosis Using Time-Domain Signal Under Road Vehicle Characteristics | V.-N. Tuyet-Doan, M. Choi, G. Park | *Electronics* (MDPI) | 2025 |

# Paper 1: A Review of OBD-II-Based Machine Learning Applications for Sustainable, Efficient, Secure, and Safe Vehicle Driving

*E. T. Michailidis, A. Panagiotopoulou, and A. Papadakis*

- A broad survey of how machine learning is applied to OBD-II sensor data across fuel optimisation, emission control, driver-behaviour profiling, vehicle health monitoring, anomaly detection, and cybersecurity.
- Covers supervised, unsupervised, deep-learning and hybrid approaches, and compares their suitability for real-time, low-latency OBD-II processing.
- Highlights open challenges: model generalisation across vehicles, scalability, computational cost, and data privacy.
- **Relevance to this project:** Confirms that OBD-II + ML for vehicle-health monitoring is an active, well-motivated research area, and frames where a real-data-only pipeline (as used here) fits among existing approaches.

# Paper 2: Driving behavior analysis and classification by vehicle OBD data using machine learning

*R. Kumar and A. Jain*

- Uses OBD-II telemetry (speed, RPM, pedal position, engine load, and other parameters) to classify driving behaviour into distinct categories based on fuel consumption, steering stability, velocity stability, and braking patterns.
- Applies SVM, AdaBoost, and Random Forest classifiers directly on time-series sensor values.
- Demonstrates that OBD-II data alone, without extra external sensors, is sufficient to build a meaningful behaviour/fault-relevant classification pipeline.
- **Relevance to this project:** Supports the choice of Random Forest as a baseline classifier and shows that OBD-II-only feature sets can achieve strong classification performance, motivating the engineered-feature approach used here.

# Paper 3: Time-Series Anomaly Detection in Automated Vehicles Using D-CNN-LSTM Autoencoder

*F. Khanmohammadi and R. Azmi*

- Proposes a D-CNN-LSTM autoencoder for detecting anomalies in time-series data from automated vehicles.
- Improves preprocessing quality using differencing and moving-standard-deviation transforms before the deep model sees the data.
- Combines convolutional feature extraction with LSTM temporal modelling inside the autoencoder to better catch low-magnitude anomalies.
- **Relevance to this project:** Directly supports the project's planned LSTM autoencoder stage — trained only on real normal sequences, flagging unusual behaviour via reconstruction error, exactly as outlined in the future-work section of this project.

# Paper 4: XGBoost Method-Based Gearbox Fault Diagnosis Using Time-Domain Signal Under Road Vehicle Characteristics

*V.-N. Tuyet-Doan, M. Choi, and G. Park*

- Proposes an XGBoost-based fault-diagnosis method for vehicle gearboxes using time-domain, statistically engineered features extracted from wheel-speed signals under real operating conditions.
- Compares XGBoost against Decision Tree and Random Forest baselines, with XGBoost consistently outperforming both.
- Shows engineered time-domain statistical features (not raw signals) are enough for strong tree-based fault classification.
- **Relevance to this project:** Validates the project's own methodology of using engineered features (Kalman-filtered signals, rolling statistics, rate-of-change, FFT peaks, ratio features) instead of raw sensor columns for the XGBoost/Random Forest benchmark.

# Work Done so far

- **Data Acquisition:** Extracted and consolidated time-series diagnostic data from the 19drivers OBD-II dataset alongside real-world KIT route files.
- **Step 1 - Preprocessing:** Cleaned European decimal formats, rectified malformed GPS coordinates, and applied Kalman filtering to smooth critical sensor noise such as RPM and Coolant Temperature.
- **Step 2 - Anomaly Detection:** We evaluated an LSTM (Long Short-Term Memory) network. Because the LSTM analyzes continuous sequences of time rather than single, isolated snapshots, it maintains the overall accuracy
- **Step 3 - Classification:** Injected synthetic anomalies (coolant overheat, MAP instability, RPM misfire) to train and benchmark Random Forest.
- **Step 4 - Behavior Profiling:** Engineered time-series features(for lstm) (speed, acceleration, engine load) and clustered the 19 distinct drivers into Low, Mid, and High intensity usage profiles using K-Means.

# Results so far

| **Application** | **ML Model Used** | **Accuracy** |
|---|---|---|
| Anomaly Detection | LSTM | 95.080% |
| Fault Classification | Random Forest, XGBoost | 94.686% |
| Driver Behaviour | K-means | 99.806% |

Each model was independently selected as the most suitable choice for its respective application based on the nature of the task and data characteristics.

# Time Plan

| Period | Activity |
|---|---|
| July, Week 3–4 | Project scope finalized; literature study and relevant vehicle datasets identified. |
| August, Week 1–2 | Final datasets collected; preprocessing, structuring and feature engineering completed. |
| August, Week 3–4 | Anomaly detection and fault classification models developed; ML models compared and best models selected. |
| September, Week 1 | **Review 1 (today)** |
| September, Week 2–4 | Driver behaviour analysis; relationship with vehicle wear studied; ML components integrated into the predictive-maintenance framework. |
| October, Week 1–4 | Dashboard/framework development; system integration, testing, validation and final documentation. |

# Conclusion

- Studied the problem of reactive, breakdown-driven vehicle maintenance and the gap in driver-accessible diagnostics.
- Finalized project objectives and completed a literature survey on anomaly detection and fault classification approaches.
- Identified suitable datasets and outlined a plan for preprocessing and model development.
- Established a clear framework for translating model outputs into plain-language insights.
- **Next steps:** dataset preprocessing, model training (anomaly detection & fault classification), and evaluation in Phase 2.

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**Thank You !!!**